Abstract

The Intelligent Hospital Queue & Appointment Optimization System is designed to automatically predict patient waiting times, estimate empirical uncertainty intervals, detect operational bottlenecks, and provide real-time decision support for hospital outpatient departments (OPDs). With the rapid growth of hospital patient inflow and unpredictable emergency cases, manual queue management leads to severe overcrowding, patient anxiety, and inefficient doctor allocation. The proposed system utilizes Discrete-Event Simulation (DES) using SimPy to model realistic patient encounters across multiple clinical departments, capturing arrival dynamics, doctor availability, triage delays, and emergency preemptions. Data preprocessing techniques, including chronological train/validation/test splitting and ColumnTransformer scaling/encoding, are combined with domain feature engineering (such as effective doctor load, traffic momentum, emergency pressure, and scheduled ratio ahead) to extract high-leverage predictive signals. Machine learning algorithms, primarily a tuned Random Forest Regressor alongside XGBoost, are deployed to predict waiting times, while department-specific quantile residual calibration provides reliable empirical 90% confidence intervals. The system incorporates a FastAPI backend integrated with a WebSockets push event bus for zero-polling real-time updates and an SQLAlchemy relational database. Role-based portals distinguish between staff operational controls (capacity management, "What-If" scenario simulation, bottleneck forecasting) and patient read-only observer views (OP key identity verification, ticket tracking). Furthermore, a continuous MLOps feedback loop tracks live performance metrics (MAE, RMSE, PSI drift) and triggers autonomous zero-downtime model retraining when performance drifts. The system provides a faster, automated, and data-driven approach for hospital operations management, significantly reducing patient wait times and administrative burden.

Keywords: Intelligent Hospital Queue System, Discrete-Event Simulation (SimPy), Random Forest Regressor, Feature Engineering, Prediction Intervals, FastAPI, WebSockets, MLOps, Role-Based Access Control, Data Visualization.

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